

CH 3: Numerical Integration

Rule	Definition	When to Use	Advantages	Disadvantages
Trapezoidal Rule	Approximates the area under a curve by dividing into straight-line segments (trapezoids).	When function is nearly linear or for rough quick approximations.	- Simple and easy - Needs less computation	- Less accurate for curved functions - Large error if function is highly non-linear
Simpson's 1/3 Rule	Approximates the curve using parabolas (degree 2 polynomial) over every 2 subintervals.	When number of intervals is even and function is smooth.	- High accuracy for smooth functions - Simple for small n	- Requires even number of intervals - Not good if function changes rapidly
Simpson's 3/8 Rule	Approximates the curve using a cubic polynomial over 3 subintervals (4 points).	When number of intervals is a multiple of 3. (i.e., 3, 6, 9,...)	- Slightly better for cubic functions - Handles 3 subinterval grouping	- Slightly complex than 1/3 rule - Not efficient if intervals not multiple of 3
Weddle's Rule	Approximates using a degree 6 polynomial over 6 subintervals (7 points).	When very high accuracy is needed and nnn is a multiple of 6.	- Very high accuracy - Good for smooth complex functions	- Tedious to apply manually - Needs exactly 6 subintervals (or multiple of 6)

🔑 Important Conditions:

Rule	Condition on number of intervals
Trapezoidal	Any number (no restriction)
Simpson's 1/3 Rule	Number of intervals must be even
Simpson's 3/8 Rule	Number of intervals must be multiple of 3
Weddle's Rule	Number of intervals must be multiple of 6

📌 Quick Memory Tip:

- Straight lines → Trapezoidal (degree 1)
- Parabola fitting → Simpson's 1/3 (degree 2 but exact for degree 3)
- Cubic fitting → Simpson's 3/8
- Degree 6 fitting → Weddle's (very accurate)